

Building a Price-Review Queue for Bangalore Villa & House Listings

How a valuation model and simple data-quality rules can turn 5,479 property listings into a focused review list of 574 (about 1 in 10), so reviewers spend time where pricing risk actually is.

Scope: independent houses and villas in Bangalore | Tools: Python (pandas, scikit-learn), Random Forest, Power BI-ready outputs | Data: 10,320 raw scraped listings, 5,479 usable records

1. The business problem

Property listing platforms and brokerages receive thousands of listings with asking prices set by owners, agents and builders. Some prices are unrealistically high or low, and the same property is often posted repeatedly. Checking every listing by hand does not scale, and unchecked pricing errors hurt buyer trust and lead quality.

Objective: design a repeatable process that estimates an expected price for each listing, flags the ones that deviate sharply, and separates genuine pricing concerns from data-quality noise.

2. The process I designed

1 - AUDIT Checked nulls, zero or negative values, extreme areas and price outliers	2 - CLEAN Extracted BHK from listing text, standardised seller types, grouped rare localities	3 - DE-DUPLICATE Built a listing fingerprint to detect the same property posted repeatedly
4 - ESTIMATE Random Forest predicts expected price; cross-validation grouped by fingerprint to avoid leakage	5 - FLAG Over-priced at +100% or more vs expected; under-priced at -50% or less	6 - TRIAGE Tag each flag with data-quality context before it reaches a reviewer

3. Headline results

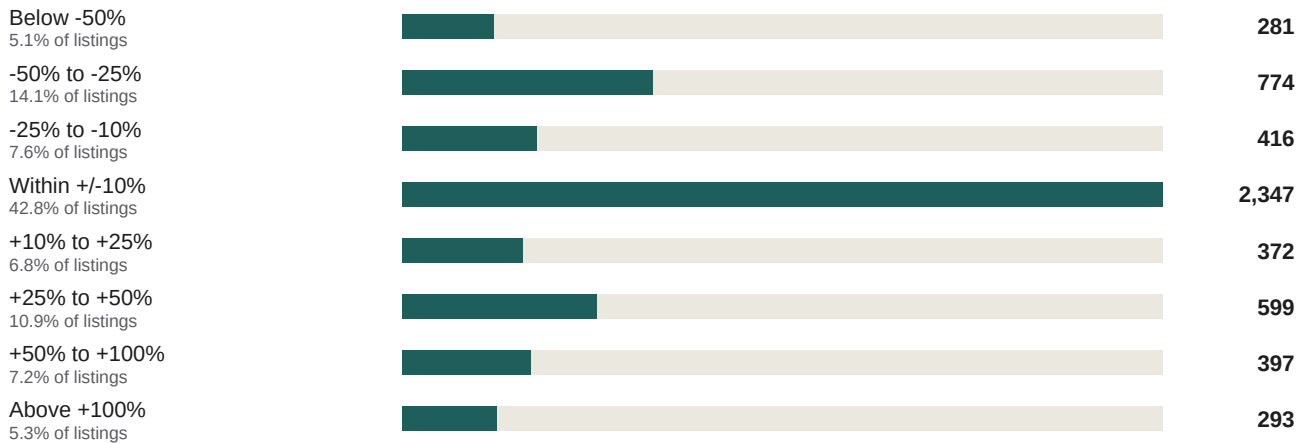
5,479 listings analysed	574 flagged for review (10.5%)	293 / 281 over-priced / under-priced
Rs 3.2 Cr median asking price	57% of listings within +/-25% of expected	71.5% are repeat postings of the same property

Why this matters: a reviewer only needs to open 10% of the inventory to cover every listing the model considers a strong pricing outlier, roughly a 90% reduction in manual checking effort for the first pass.

4. What the data shows

How far asking prices sit from the expected price

Number of listings by deviation band. Most cluster near expected value; the tails form the review queue.



Flag rate by seller type

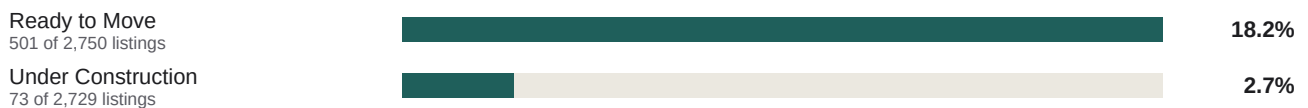
Share of each seller group's listings that were flagged.



Owner-posted listings are flagged about 3x as often as agent-posted ones (16.4% vs 5.4%). A practical rule: apply extra price validation when owners submit listings.

Flag rate by construction status

Ready-to-move properties carry far more pricing risk than under-construction ones.



Ready-to-move listings are flagged 18.2% of the time versus 2.7% for under-construction. One plausible reason is that under-construction prices follow builder price lists and so vary less.

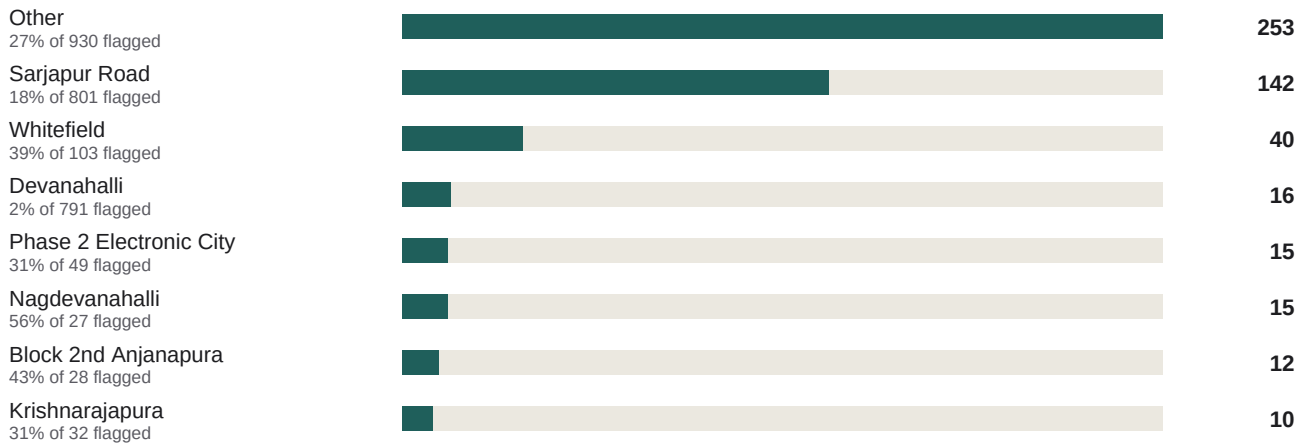
Flag rate by property type

38 listings with no property type recorded are excluded from this chart.



Where the flags concentrate: top 8 localities

Flagged listings per locality. Three areas account for 435 of 574 flags (76%).



The "Other" bucket (rare localities grouped together), Sarjapur Road and Whitefield drive most flags, so review capacity can be allocated by area.

Flag rate by data-quality status

Listings with an extreme built-up area (for example 64,000 sq ft) are flagged far more often than clean records.



Extreme-area records are flagged 41% of the time, so an area check at submission would remove many false alarms before any model runs. Repeat postings are flagged least (3.1%), likely because near-identical postings have similar prices.

5. Recommended review workflow

Tier	Trigger	Action
1 - Auto-hold	Extreme-area records that are also flagged (91 listings)	Return to submitter to correct area/price before publishing
2 - Priority review	Flagged listings with no data-quality issue (363 listings)	Human reviewer checks against comparable listings
3 - Duplicate clean-up	Flagged repeat postings (120 listings)	Merge with the original posting and keep one price
4 - Monitor	Unflagged listings	No action; re-score on next data refresh

Ongoing controls: monthly re-scoring, tracking flag rate by seller type and locality, and using reviewer decisions as feedback to tune the thresholds.

6. Limitations

- Model accuracy is modest.** Grouped cross-validation R2 is 0.26 and mean absolute error is about Rs 94.7 lakh. The data lacks property age, finish quality and exact micro-location, so the model ranks relative outliers rather than quoting a price.
- Segment scope.** The source contains only villas and residential houses, with no apartments. Findings apply to that segment.

- **Heavy duplication.** About 72% of records are repeat postings, so far fewer truly independent properties exist than the row count suggests (1,559 non-repeated records plus repeat groups).
- **Flags are review candidates, not verdicts.** A flagged listing may be priced deliberately high or low for reasons the data cannot see.
- **Uneven flag coverage.** Three localities made up entirely of repeat postings (Thirupalya, Bidaraguppe, Hennur Gardens; 1,962 listings) produced no flags and should be sampled manually.

7. Skills demonstrated

Process design | Data quality auditing | Python and pandas | Machine learning (Random Forest) |
Cross-validation without leakage | Exception-based triage | Power BI | Stakeholder-ready reporting

Source data: Bangalore real-estate listings (10,320 raw records, 5,479 modelled). All figures are computed from the project's exported output files.